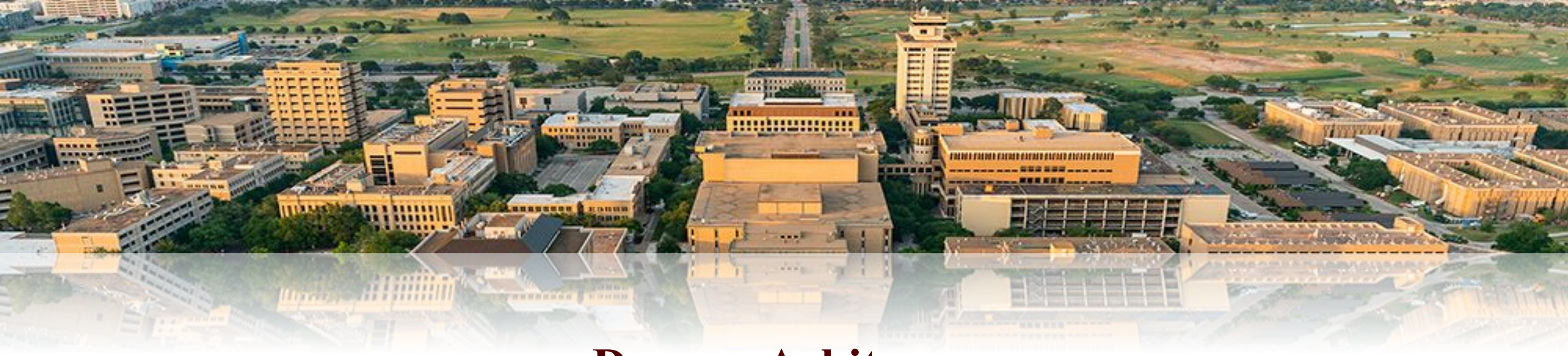
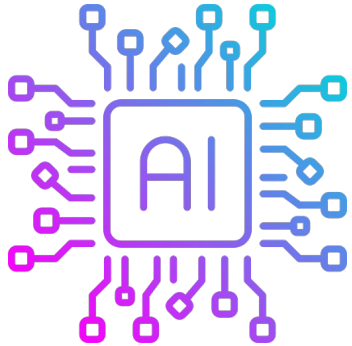


DamageArbiter:
A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery

Yifan Yang, Ph.D. Student
Advisor: Dr. Lei Zou (lzou@tamu.edu)
Department of Geography, Texas A&M University
GEAR Lab Website: https://www.geoearlab.com

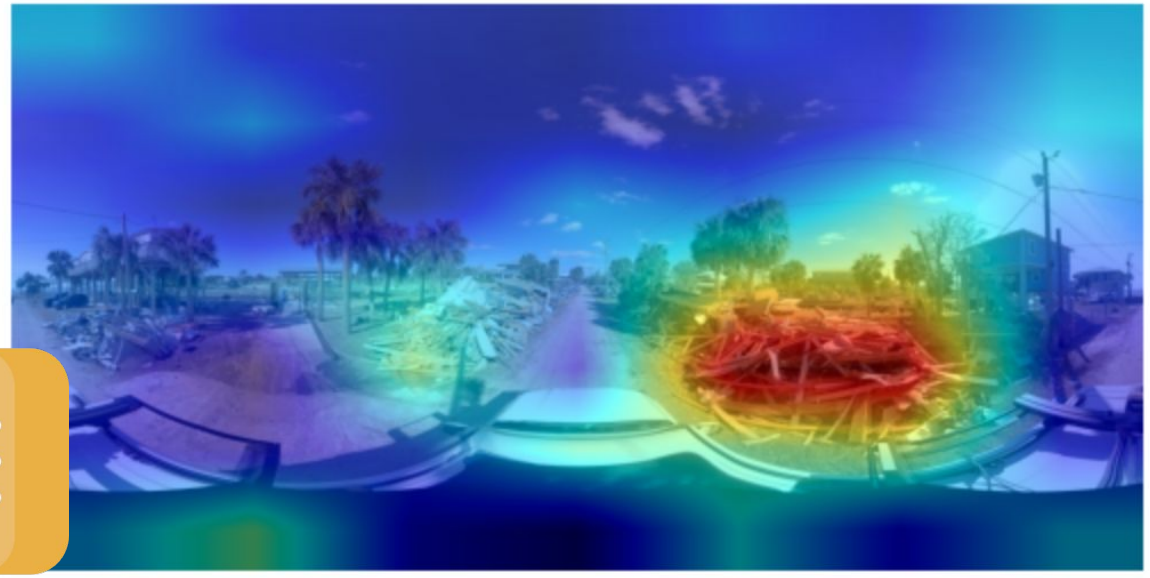
Introduction

Post-Hurricane Street-View Imagery



Computer Vision–Based Deep Learning Models

Grad-CAM for Fine-Tuned Swin-Small



Severe

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

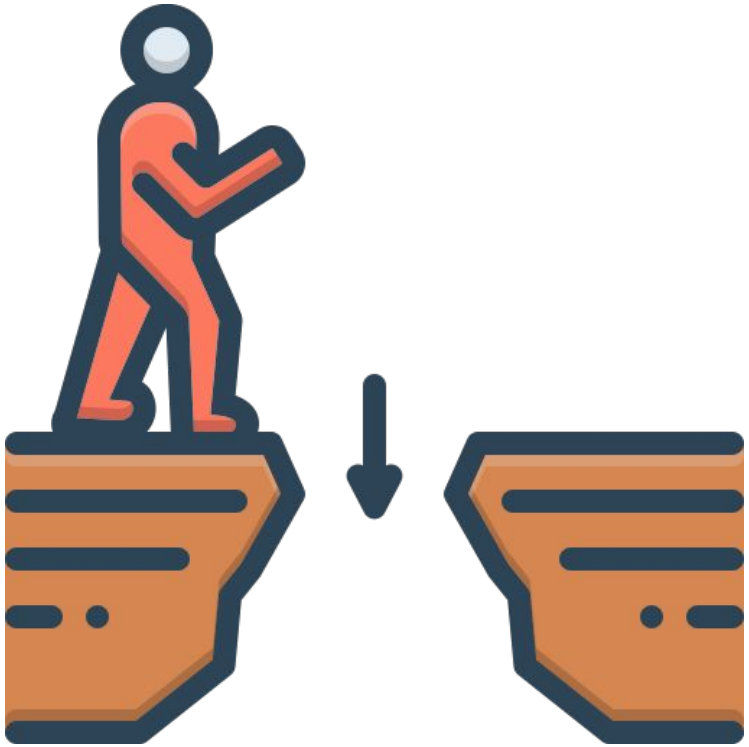
Research Gaps

Research Gaps in Street-View–Based Disaster Assessment

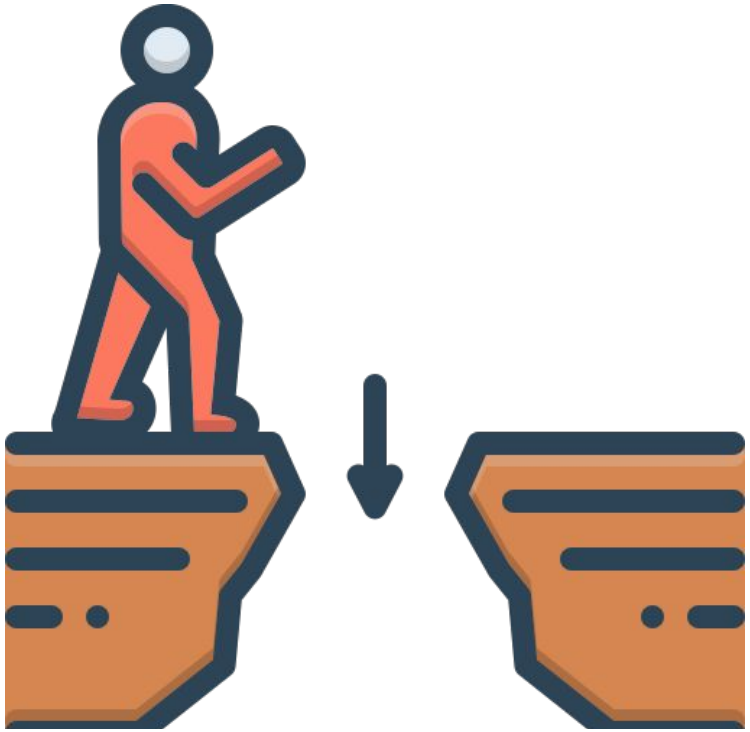
Limitations of Traditional Deep Vision Models

- Heavy reliance on manual annotations
- Coarse damage categories (3–4 levels only)
- Limited semantic reasoning capability
- Black-box predictions with weak interpretability
- Overconfidence under uncertainty

Research Questions



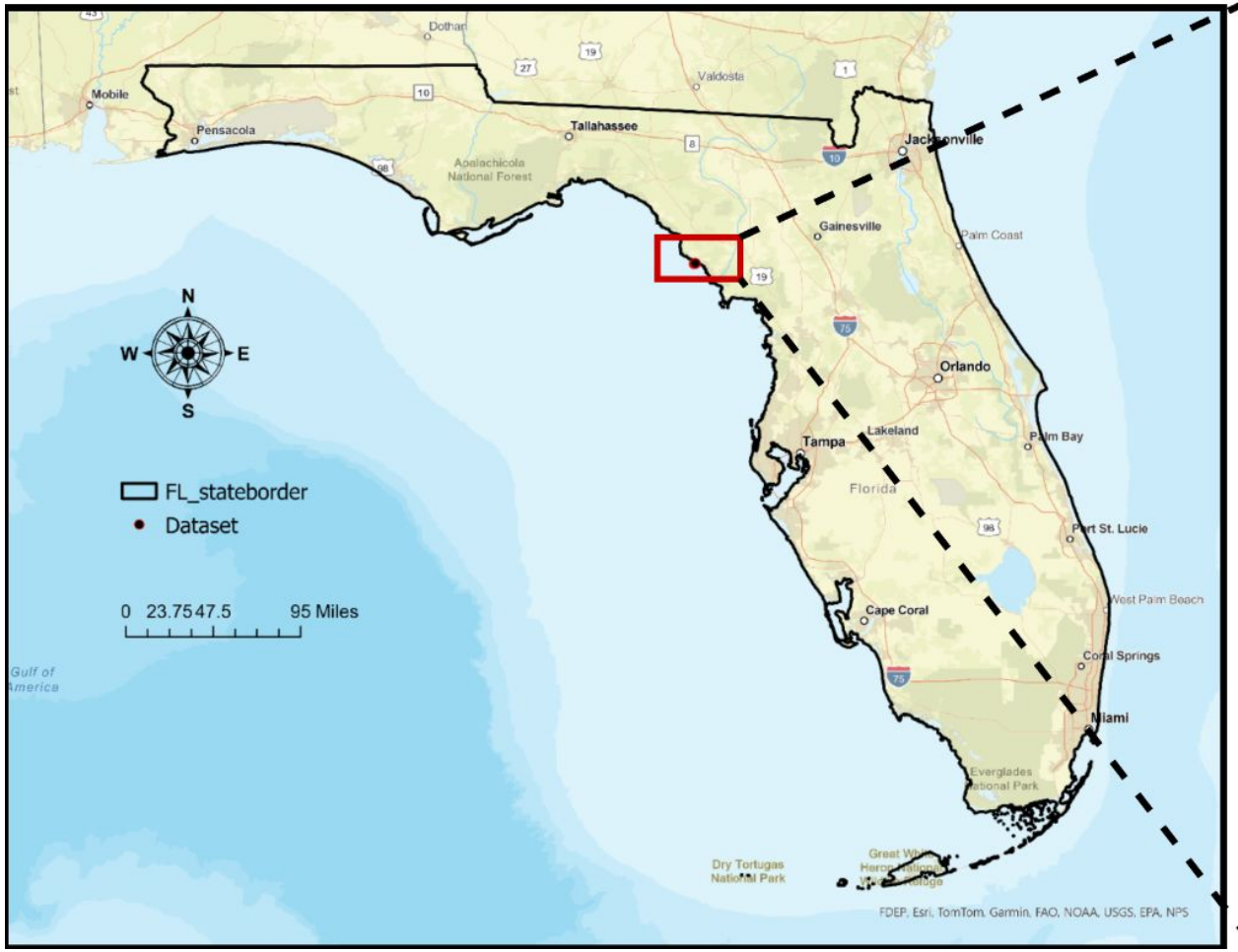
RQ1: How can we improve classification accuracy for rapid disaster assessment?

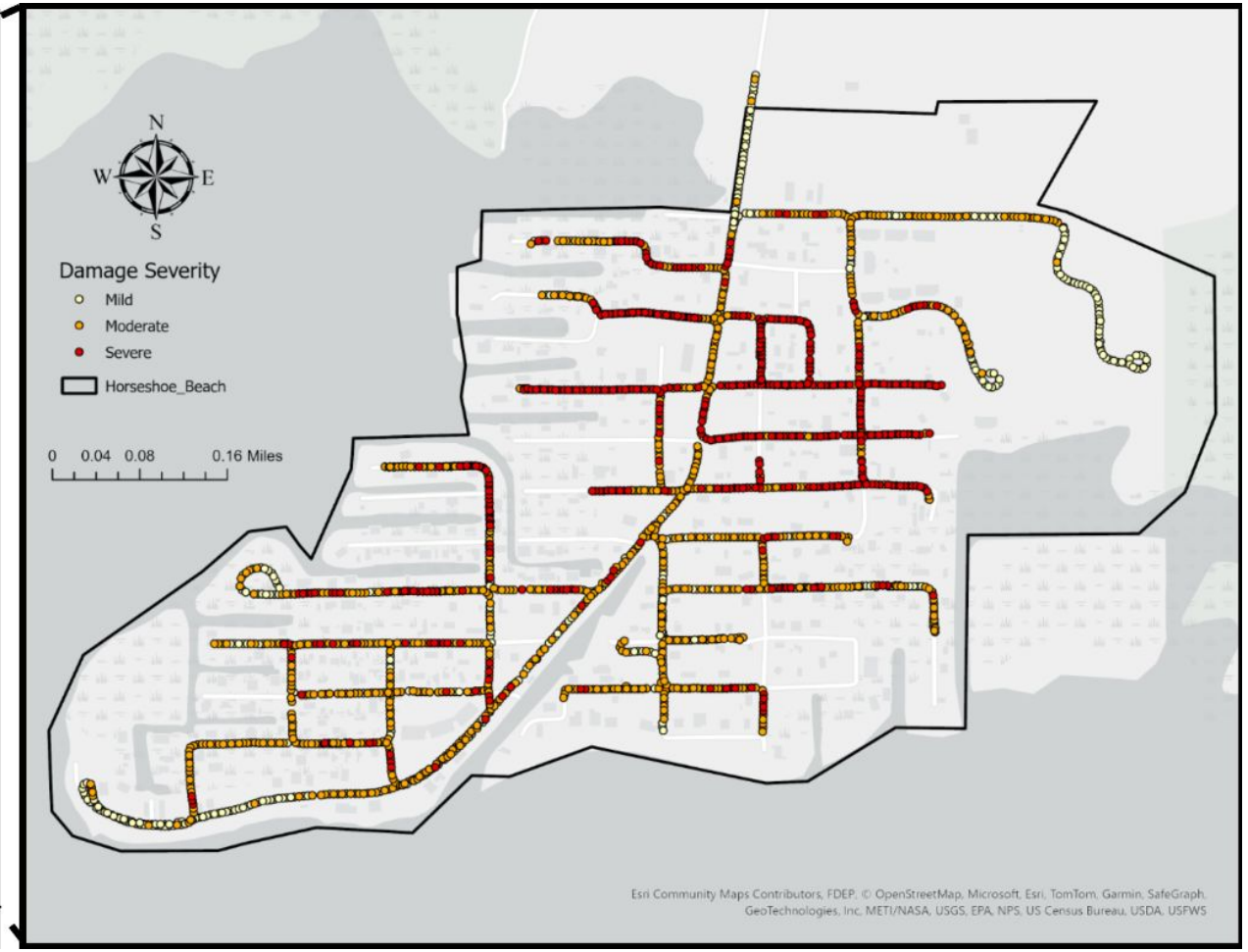


RQ2: How can we evaluate and mitigate model overconfidence in high-stakes scenarios?

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Dataset & Study Area





We leveraged the Hurricane Damage Street-View Imagery dataset constructed by Yang et al. (2025). The dataset contains 2,556 post-disaster street view images. Each post-disaster image was manually classified into three damage severity levels: mild (658 images), moderate (1,196 images), and severe (702 images).

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Methodology

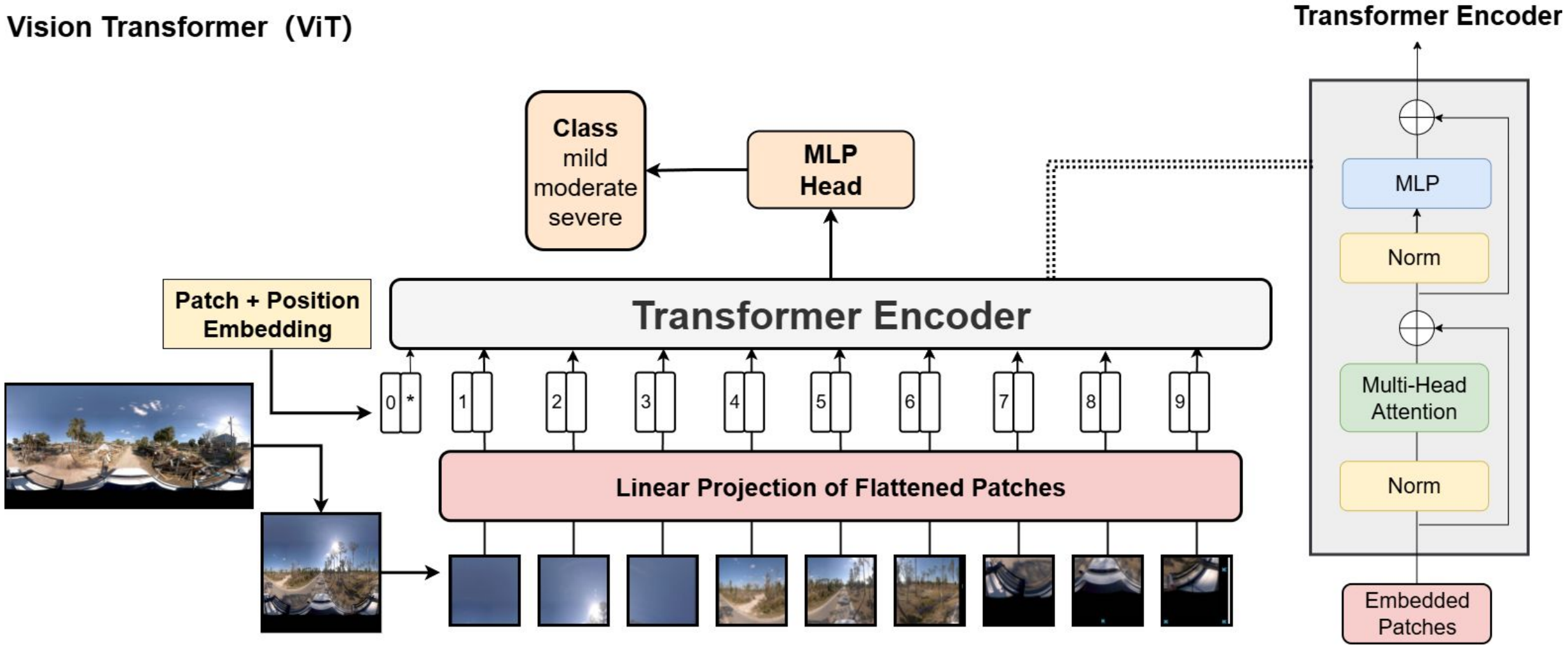


Figure. Vision Transformer (ViT) Framework for Disaster Assessment

Methodology

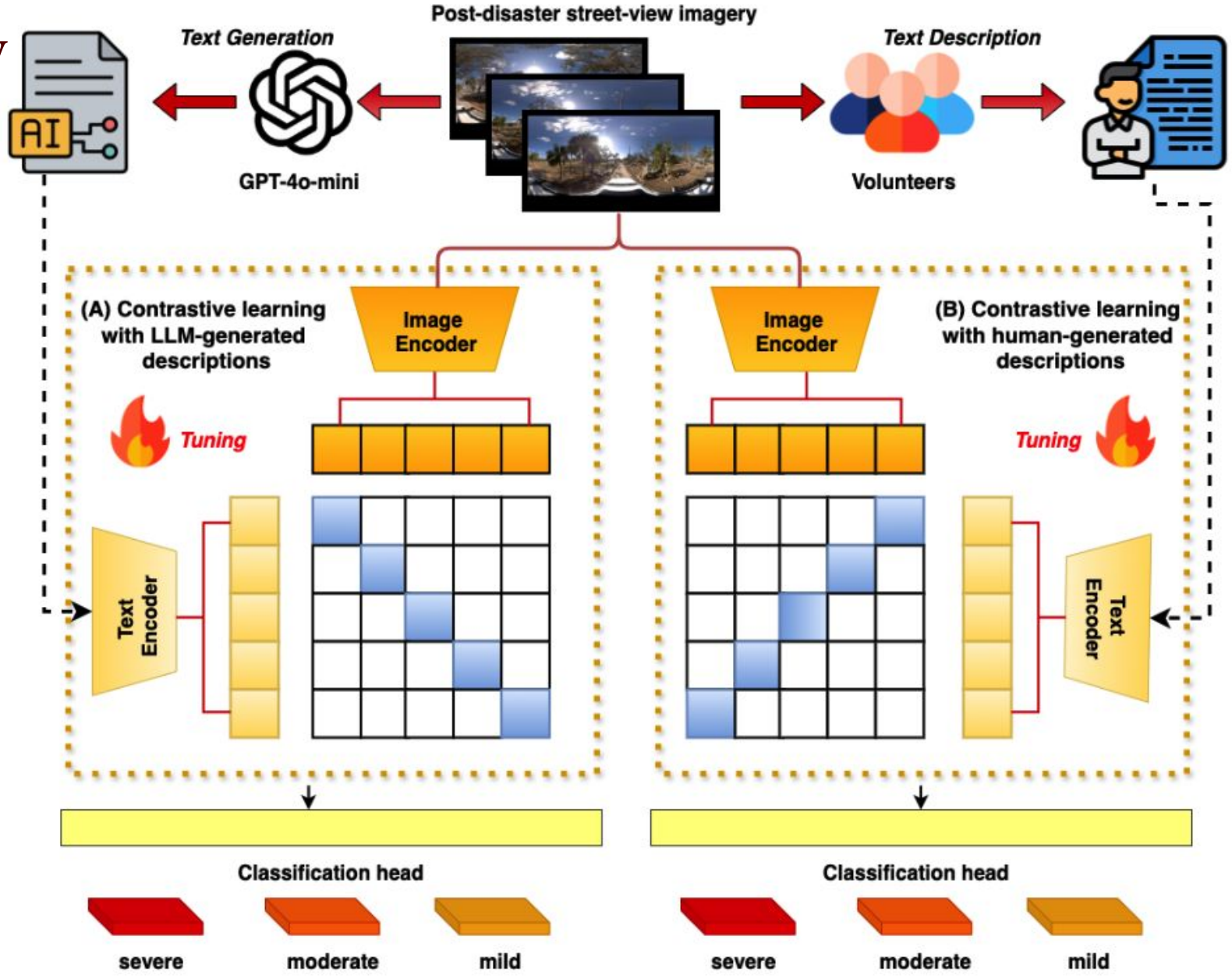


Figure. Contrastive Learning for Disaster Assessment with Human vs. LLM-Generated Descriptions.

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Examples of Disaster Severity Categories with Human- and LLM-Generated Descriptions

class0 mild

No buildings in the area were damaged by the hurricane, and the trees appear to have sustained only mild damage. This is truly a relief. The vegetation is lush, the trees are tall, and the damage from the hurricane was minimal. I'd love to live here now, as it's only slightly damaged and looks very livable.

Sagging power lines are visible alongside a clear, dry road. Many trees show defoliation and some branch damage, but no large fallen trees or building debris are present. The overall scene suggests mild environmental impact, with infrastructure largely intact.

class1 moderate

The damage level is moderate, with loose building materials and a small amount of debris suggesting recent construction or minor disrepair. Despite the somewhat cluttered appearance of scattered items, the water features and elevated residences make this area appear to be an attractive residential area.

Scattered wooden debris and loose utility lines on a power pole indicate moderate damage along this coastal dirt road. While buildings appear structurally sound, the presence of scattered planks and beams, alongside the compromised power infrastructure, suggests the area experienced some impact, but not widespread destruction or flooding.

class2 severe

The atmosphere of the image is chaotic, with massive piles of debris covering large parts of the area, giving a strong sense of destruction. The damage appears extensive, with shattered building materials strewn all over, although some elevated houses remain intact in the background.

The street is heavily impacted by extensive building debris, forming large piles along both sides, almost completely blocking passage. This significant accumulation of scattered materials and collapsed fragments suggests severe damage to structures in the vicinity, though visible infrastructure like power lines appears mostly intact.

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Contrastive Language-Image Pretraining (CLIP) model

TEXAS A&M UNIVERSITY

1) **Visual and textual features are extracted using a pre-trained CLIP model, with the image processed through a visual encoder (ViT-B/32) and the text through a transformer-based text encoder.**

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\text{sim}(x_i, y_i)/\tau\right)}{\sum_{j=1}^{N}\exp\left(\text{sim}(x_i, y_j)/\tau\right)}$$

2) **A lightweight linear classification head is trained to predict one of three disaster severity levels (mild, moderate, or severe) using the standard cross-entropy loss**

$$\mathcal{L}_{\text{cls}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\log\hat{y}_{ic} \qquad \mathcal{L}_{\text{total}} = \lambda\mathcal{L}_{\text{contrast}} + (1-\lambda)\mathcal{L}_{\text{cls}}$$

3) **CLIPScore quantifies how well an image–text pair aligns within the shared multimodal embedding space learned during CLIP's training.**

$$\text{CLIPScore}(I, T) = \cos(\mathbf{v}_I, \mathbf{v}_T) = \frac{\mathbf{v}_I \cdot \mathbf{v}_T}{\|\mathbf{v}_I\|\|\mathbf{v}_T\|}$$

EAR Lab
Explore & Resolve

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Understanding Model Reliability in Disaster Damage Assessment

TEXAS A&M UNIVERSITY

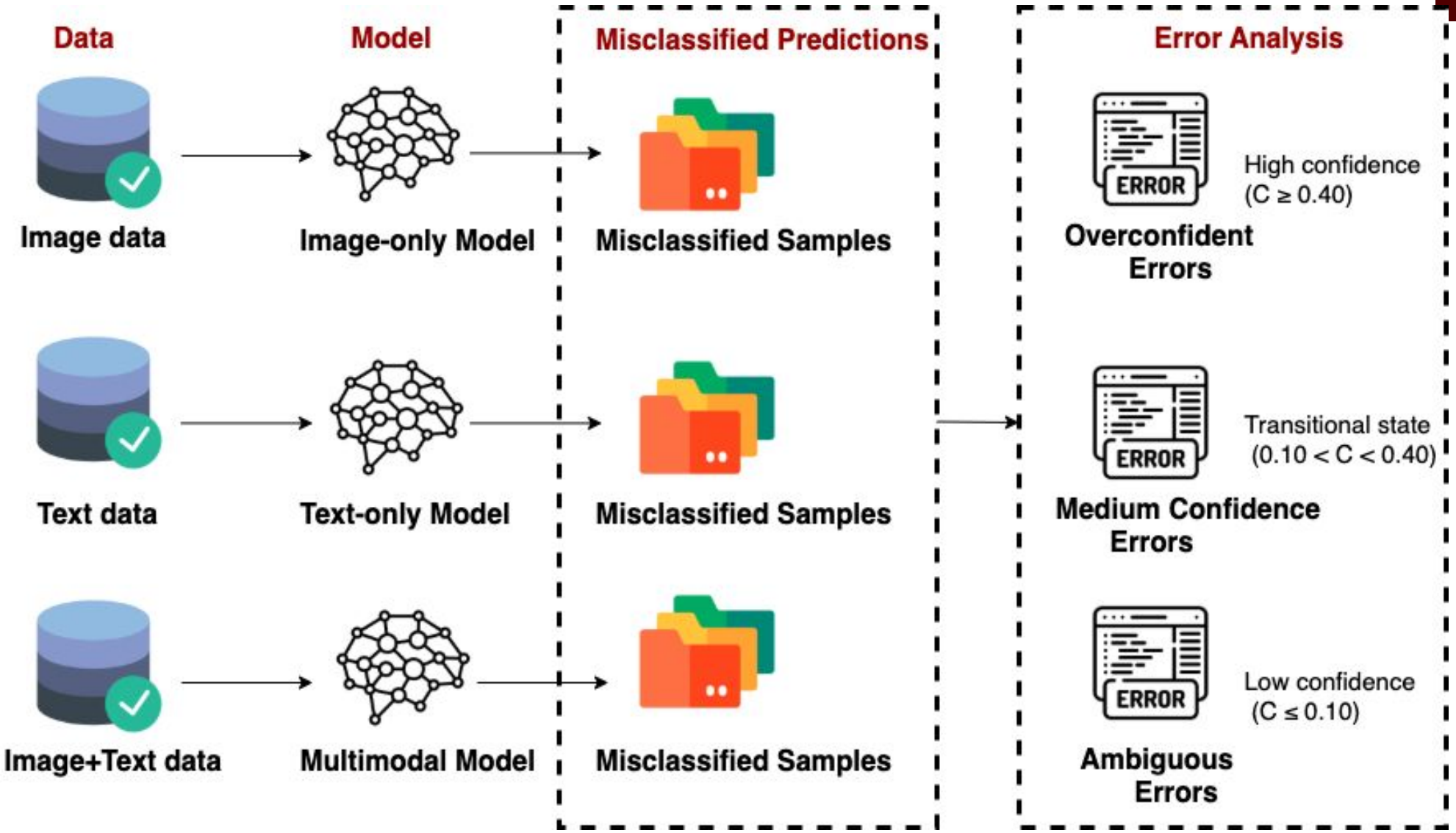


Figure. Framework of Confidence-Based Error Analysis in Disaster Assessment Models.

EAR Lab
Explore & Resolve

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Results for Decision Confidence and Ambiguity Analysis

TEXAS A&M UNIVERSITY

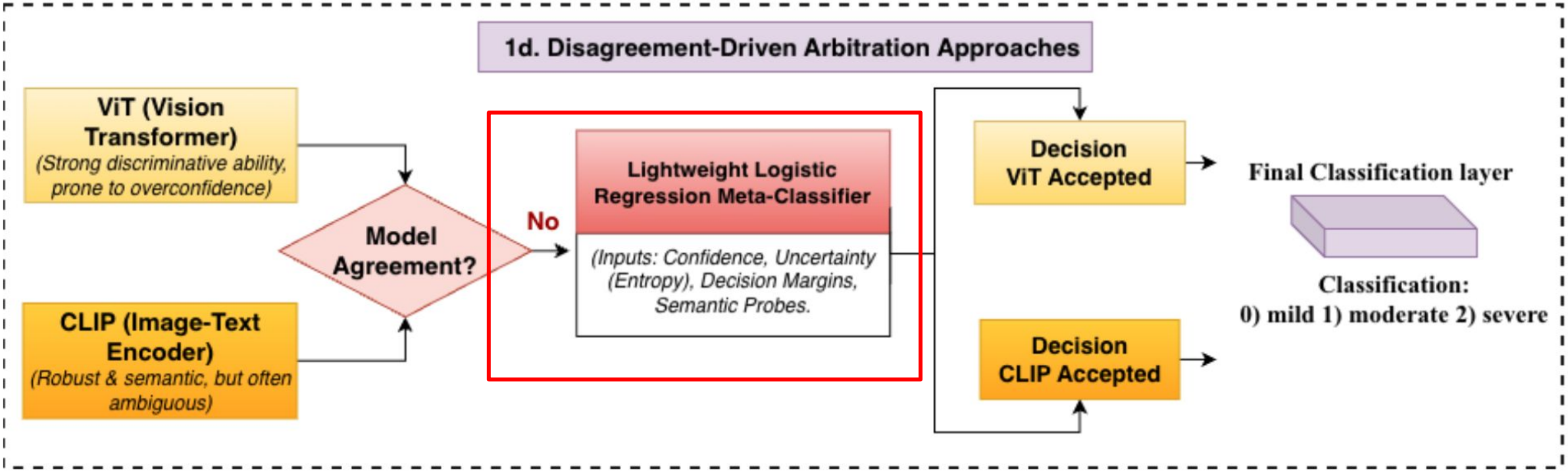
Table 2. Confidence- and Severity-Based Misclassification Profiling of Image-, Text-, and Multimodal Models. Overconfident, Medium, and Ambiguous represent the misclassification distribution at different confidence levels; Mild Mis, Moderate Mis, and Severe Mis represent the misclassification proportions at different disaster levels.

Model	Modality	Overconfident	Medium	Ambiguous	Mild Mis	Moderate Mis	Severe Mis
ViT-B/16	Post Image	64.86%	27.14%	8.00%	40.43%	35.00%	24.57%
ViT-B/32	Post Image	70.58%	21.04%	8.38%	47.26%	27.90%	24.85%
Text Encoder (LLM)	Text Only	47.78%	37.71%	14.51%	38.14%	38.67%	23.20%
Text Encoder (Human)	Text Only	53.27%	33.65%	13.08%	30.20%	39.95%	29.85%
CLIP (LLM)	Image+Text	0.00%	55.18%	44.82%	39.73%	24.25%	21.91%
CLIP (Human)	Image+Text	0.00%	66.82%	33.18%	40.49%	19.06%	24.89%

EAR Lab
Explore & Resolve

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Disagreement-Driven Arbitration Framework



This mechanism enables refined judgment on disagreement cases:
when ViT is overconfident, the robustness of CLIP prevents misjudgments;
and when CLIP is ambiguous, ViT's strong discriminative power provides clearer decisions.
Ultimately, all prediction results are fed into the classification layer, which outputs three disaster severity categories (mild, moderate, and severe).

Comparison of Disaster Assessment Modeling

TEXAS A&M UNIVERSITY

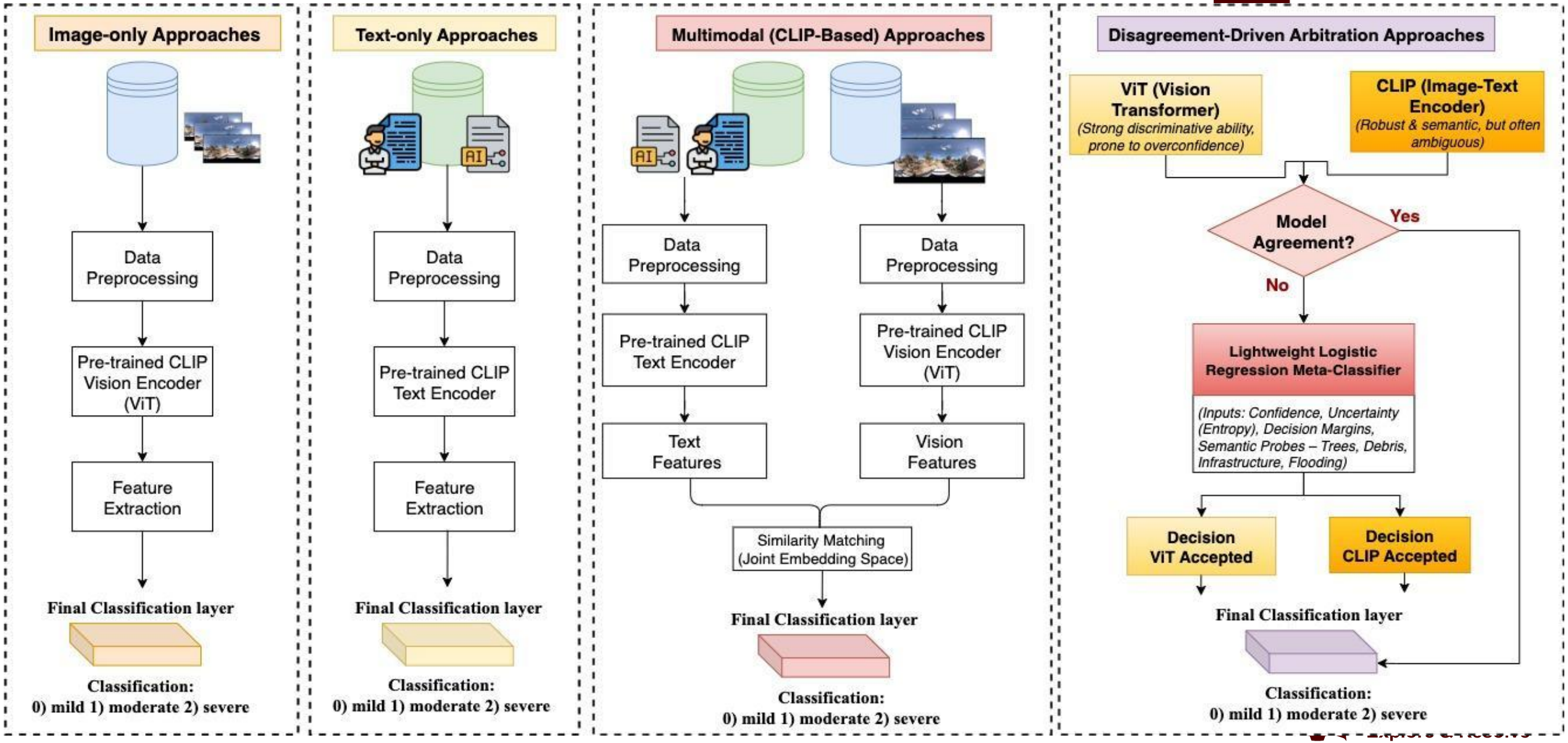


Figure . Overall Framework of Disaster Assessment Approaches.

Results for Multimodal Contrastive Representation Learning

TEXAS A&M UNIVERSITY

Table1. Comparative Evaluation of Image-only, Text-only, and Multimodal Models on Disaster Severity Classification.

Model	Modality	Accuracy	Recall	Precision	SW-F1	MCC	Clipscore
ViT-B/16	Image Only	0.7261	0.7261	0.7320	0.7240	0.5653	-
ViT-B/32	Image Only	0.7433	0.7433	0.7597	0.7386	0.5947	-
Text Encoder (LLM)	Text Only	0.6307	0.6307	0.6331	0.6261	0.4126	-
Text Encoder (Human)	Text Only	0.6710	0.6710	0.6779	0.6715	0.4788	-
CLIP (LLM)	Image+Text	0.7238	0.7238	0.7249	0.7229	0.5644	0.2467
CLIP (Human)	Image+Text	0.7422	0.7422	0.7504	0.7403	0.5915	**0.2701**
ViT-B/32 + CLIP(LLM)	Image+Text	**0.8279**	**0.8065**	**0.8555**	**0.8261**	**0.7305**	

EAR Lab
Explore & Resolve

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com

Ablation Study

Table 3. Ablation study of logistic regression arbitration with different feature configurations.

Arbitration Setting	Features Used	Accuracy	Macro-F1
LOGREG_conf [τ=0.35]	Confidence only	0.976	0.965
LOGREG_conf+unc [τ=0.40]	Confidence + Uncertainty	0.940	0.932
LOGREG_conf+unc+probe [τ=0.45]	Confidence + Uncertainty + Semantic Probes	0.940	0.932
LOGREG_probe_only [τ=0.50]	Probes only	0.607	0.560

Decision Logic:

(1) Trust ViT when it is significantly more confident than CLIP

(2) Trust CLIP when it is more confident or ViT is uncertain

(3) Select the relatively more reliable prediction

Future Directions & Takeaways

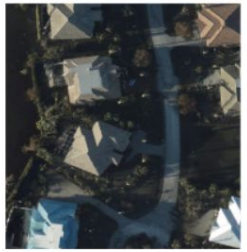
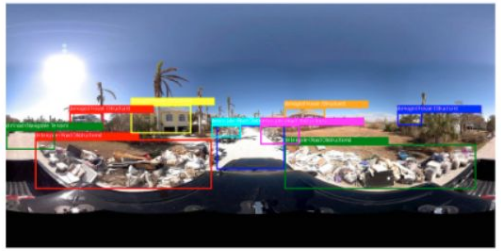
TEXAS A&M UNIVERSITY

Autonomous GeoAI

- Adaptive multimodal reasoning
- Dynamic conflict resolution
- Real-time spatial intelligence

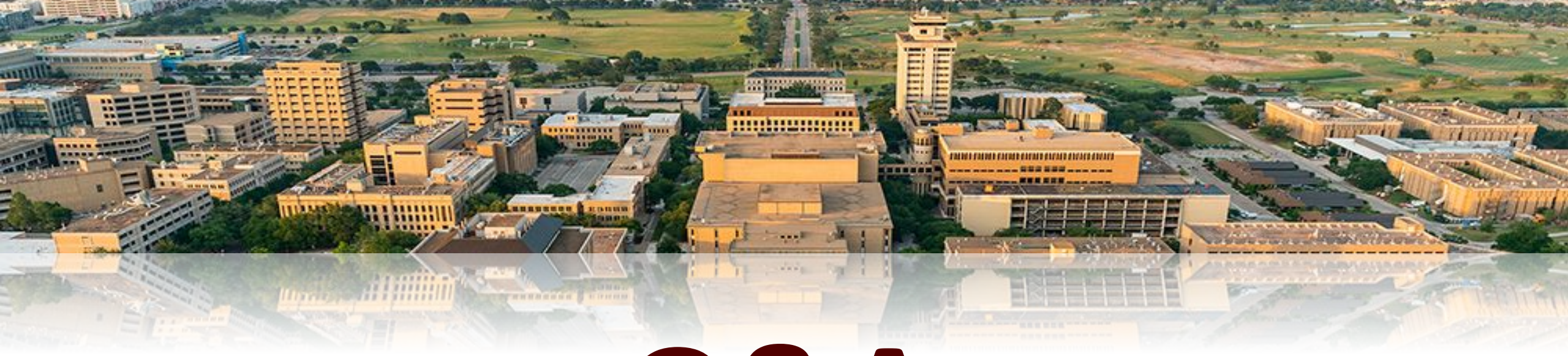
Responsible GeoAI

- Reliability before accuracy
- Overconfidence mitigation
- Transparent, trust-aware inference

Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building
 Using Gemini-3-Pro for object recognition	The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. **Reasoning for Gemini-3-Pro-Disaster Assessment**			
	Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. **Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion**			

EAR Lab
Explore & Resolve

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com



Q&A

AAG

Welcome any comments/questions

yyang295@tamu.edu

EAR Lab
Explore & Resolve